



Ordered SCOOP Peptides

Comparison and Differences



From the hierarchical clustering, we can see 3 main families:

Family 1 – AVDH SYS-like

AVDHSSSPRHGQR
AVDHSYSTRNGQR
AVDHSVSARAGQR
AVDRSYSPRGGQR
GVDHSYSQRAGQR

These share:

- Positions 1–4: AVDH/GVDH motif
- C-terminus often GQR

Family 2 – RVGRSKS-like

RVGRSKSLRGPQK
RVHRSSSKRGPQK
HVRSSRSQIGIQK

These share:

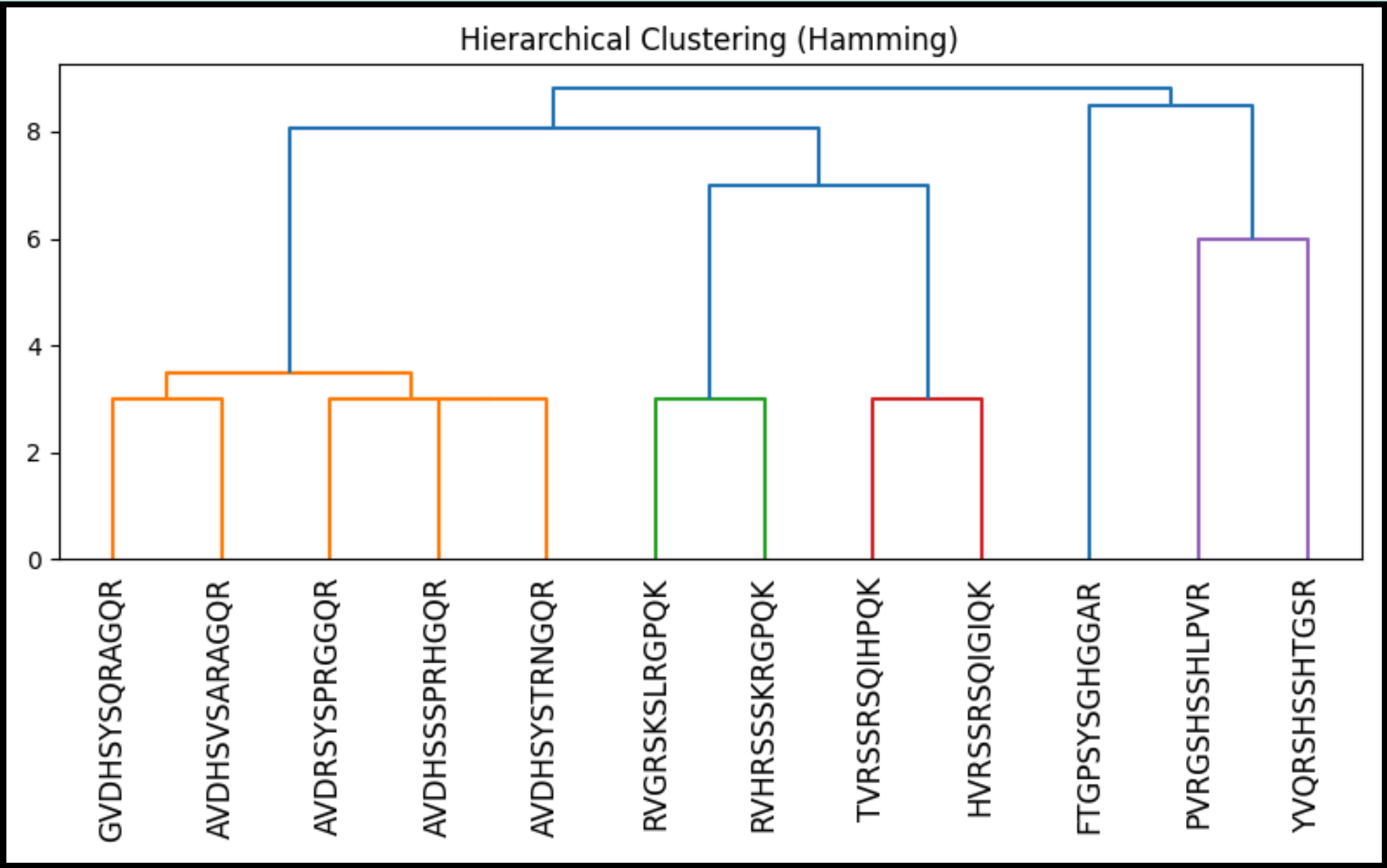
- Strong enrichment in Arg/Lys
- Highly basic peptides
- Often QK ending

Family 3 – WT-like

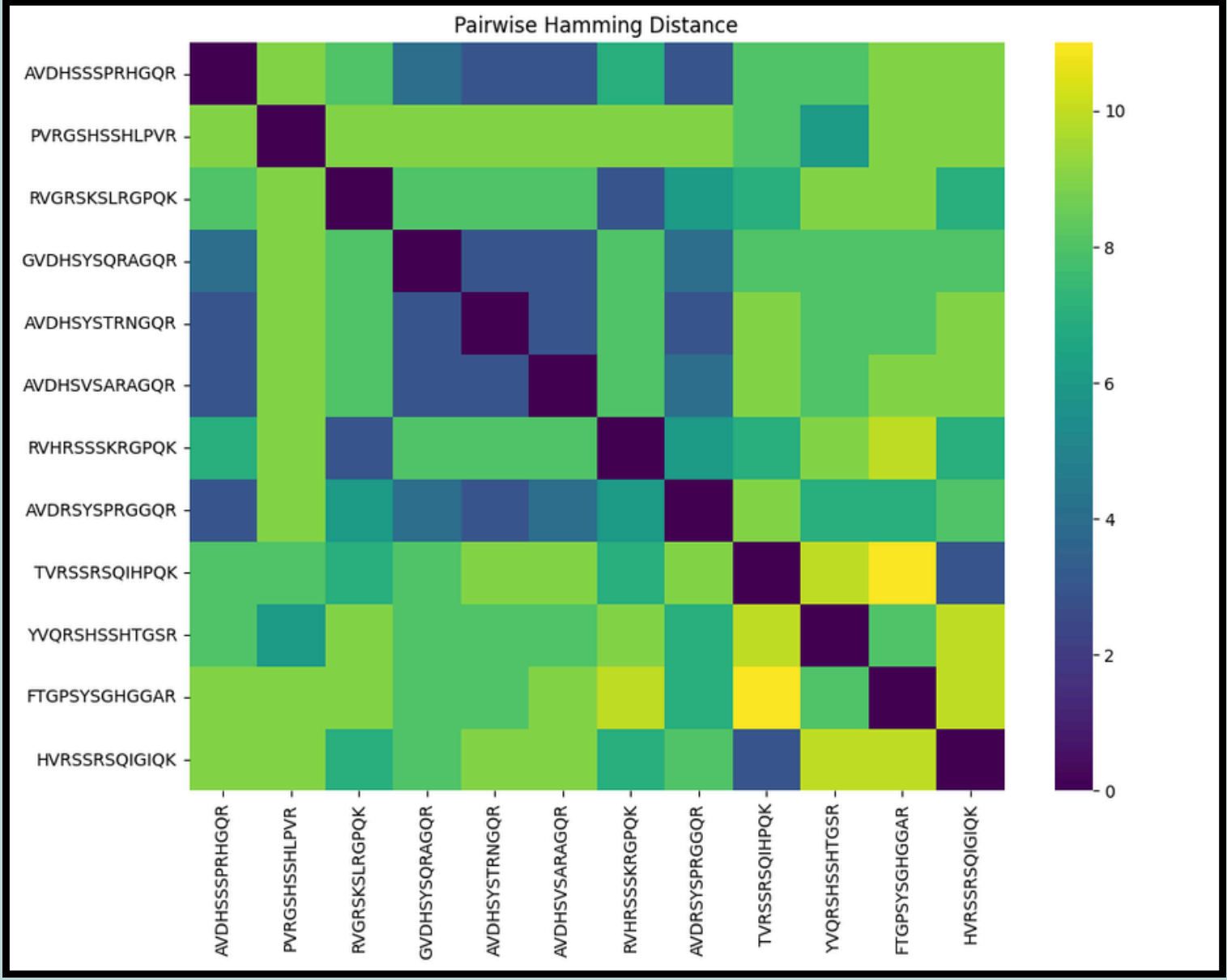
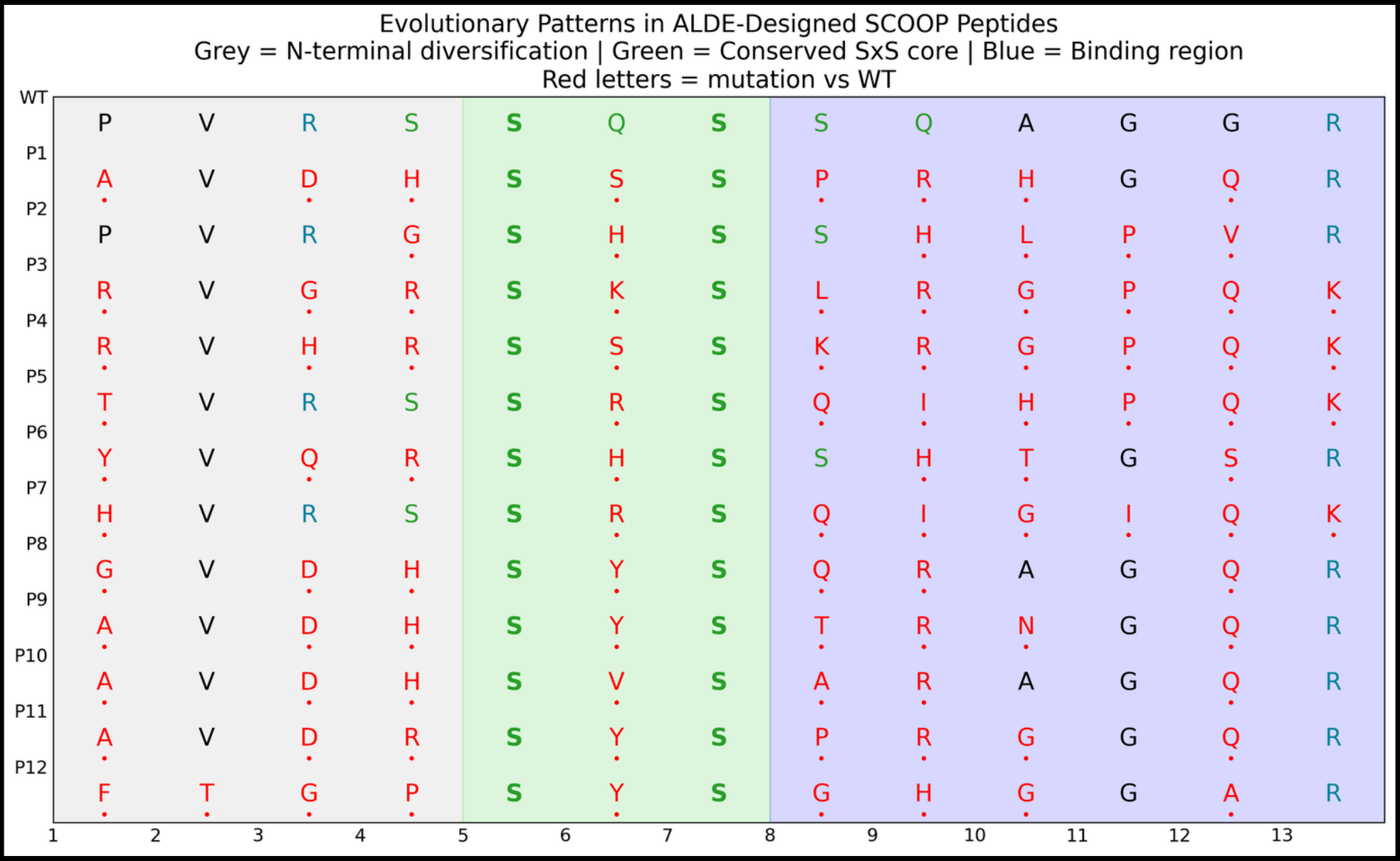
PVRGSHSSHLPVR
YVQRSHSSHTGSR

- These are closer to WT architecture but with scattered substitutions.

Clustering of ALDE SCOOPs

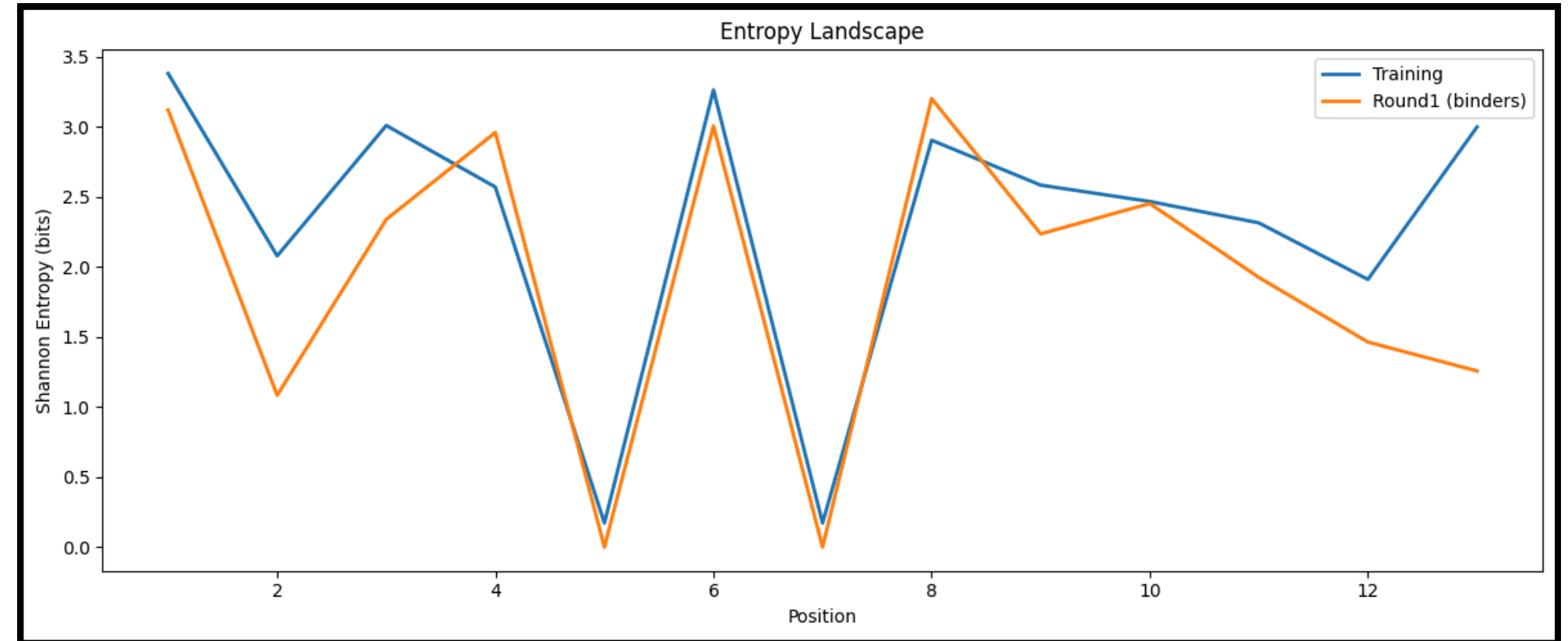


MSA & Pairwise Distances



Entropy calculation

Shannon entropy at each position measures residue diversity. High entropy means many different amino acids appear there. Low entropy means conservation.



Negative Δ entropy = ALDE reduced diversity.

Positive Δ entropy = ALDE increased diversity.

Decreased entropy indicates convergence at functional hotspots.

